signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

signal.sci

```
f = (0:N-1)*(fs/N);

// Positive-frequency spectrum
idx = 1:(N/2+1);
f_pos = f(idx);
Xf_pos = Xf(idx);

// 4. Plotting
scf(0);
clf();

// Continuous signal
subplot(2,1,1);
plot(t, x);
xgrid();
xtitle("Continuous vs Sampled Signal", "Time (s)", "Amplitude");

// Overlay sampled points
plot(n, xs, "ro");

// Frequency spectrum
subplot(2,1,2);
plot(f_pos, Xf_pos);
xgrid();
xtitle("Spectrum of Sampled Signal (Aliasing Effect)", ..
       "Frequency (Hz)", "Magnitude");

// Set frequency-axis limits
a = gca();
a.data_bounds = [0, 0; fs/2, max(Xf_pos)];

```

